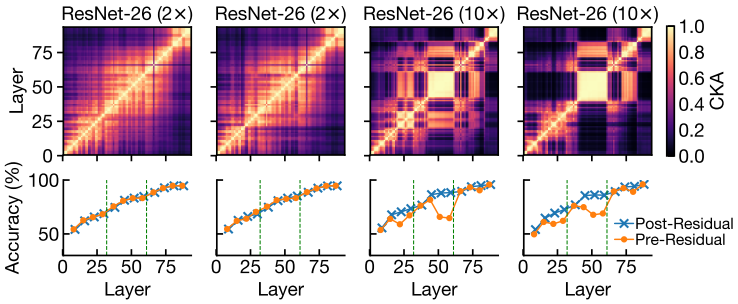

Figure 4: Linear probe accuracy. Top: CKA between layers of individual ResNet models, for different architectures and initializations. Bottom: Accuracy of linear probes for each of the layers before (orange) and after (blue) the residual connections.

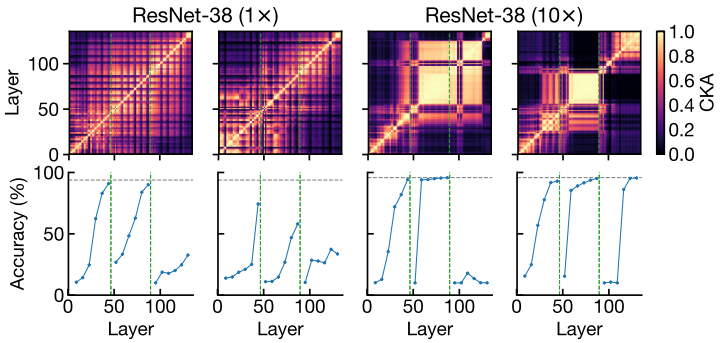

Figure 5: Effect of deleting blocks on accuracy for models with and without block structure. Blue lines show the effect of deleting blocks backwards one-by-one within each ResNet stage. (Note the plateau at the block structure.) Vertical green lines reflect boundaries between ResNet stages. Horizontal gray line reflects accuracy of the full model.

nearly eliminates the block structure from the CKA heatmaps (bottom right). A full picture of how this process impacts models of increasing depth and width can be found in Appendix Figure D.6.

In contrast, for models that do not contain the block structure, we find that cosine similarity of the first principal components across all pairs of layers bears little resemblance to the representation similarity structure measured by CKA, and the fractions of variance explained by the first principal components across all layers are relatively small (see Appendix Figure D.7). Together these results demonstrate that the block structure arises from preserving and propagating the first principal component across its constituent layers.

Although layers inside the block structure have representations with high CKA and similar first principal components, each layer nonetheless computes a nonlinear transformation of its input. Appendix Figure D.8 shows that the sparsity of ReLU activations inside and outside of the block structure is similar. In particular, ReLU activations in the block structure are sometimes in the linear regime and sometimes in the saturating regime, just like activations elsewhere in the network.

5.2 Linear Probes and Collapsing the Block Structure

With the insight that the block structure is preserving key components of the representations, we next investigate how these preserved representations impact task performance throughout the network, and whether the block structure can be collapsed in a way that minimally affects performance.

In Figure 4, we train a linear probe (Alain & Bengio, 2016) for each layer of the network, which maps from the layer representation to the output classes. In models without the block structure (first 2 panes), we see a monotonic increase in accuracy throughout the network, but in models with the block structure (last 2 panes), linear probe accuracy shows little improvement inside the block structure. Comparing the accuracies of probes for layers pre- and post-residual connections, we find that these connections play an important role in preserving representations in the block structure.

Informed by these results, we proceed to pruning blocks one-by-one from the end of each residual stage, while keeping the residual connections intact, and find that there is little impact on test accuracy when blocks are dropped from the middle stage (Figure 5), unlike what happens in models without block structure. When compared across different seeds, the magnitude of the drop in accuracy appears to be connected to the size and the clarity of the block structure present. This result suggests that block structure could be an indication of redundant modules in model design, and that the similarity of its constituent layer representations could be leveraged for model compression.

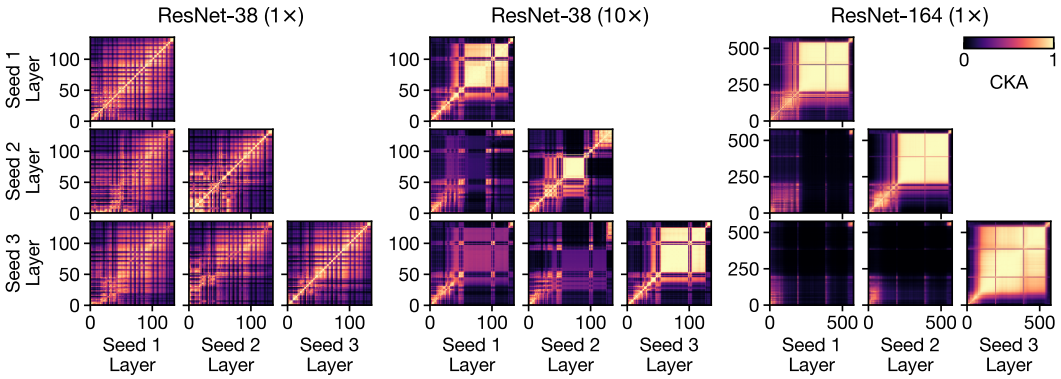

Figure 6: Representations within "block structure" differ across initializations. Each group of plots shows CKA between layers of models with the same architecture but different initializations (off the diagonal) or within a single model (on the diagonal). For narrow, shallow models such as ResNet-38 (1×), there is no block structure, and CKA across initializations closely resembles CKA within a single model. For wider (middle) and deeper (right) models, representations within the block structure are unique to each model.

6 Depth and Width Effects on Representations Across Models

The results of the previous sections help characterize effects of varying depth and width on a (single) model's internal representations, specifically, the emergence of the block structure with increased capacity, and its impacts on how representations are propagated through the network. With these insights, we next look at how depth and width affect the hidden representations *across* models. Concretely, are learned representations similar across models of different architectures and different random initializations? How is this affected as model capacity is changed?

We begin by studying the variations in representations across different training runs of the same model architecture. Figure 6 illustrates CKA heatmaps for a smaller model (left), wide model (middle) and deep model (right), trained from random initializations. The smaller model does not have the block structure, and representations across seeds (off diagonal plots) exhibit the same grid-like similarity structure as within a single model. The wide and deep models show block structure in all their seeds (as seen in plots along the diagonal), and comparisons across seeds (off-diagonal plots) show that while layers not in the block structure exhibit some similarity, the block structure representations are highly dissimilar across models.

Appendix Figure E.1 shows results of comparing CKA *across* different architectures, controlled for accuracy. Wide and deep models without the block structure do exhibit representation similarity with each other, with corresponding layers broadly being of the same *proportional* depth in the model. However, similar to what we observe in Figure 6, the block structure representations remain unique to each model.

7 Depth, Width and Effects on Model Predictions

To conclude our investigation on the effects of depth and width, we turn to understanding how the characteristic properties of internal representations discussed in the previous sections influence the outputs of the model. How diverse are the predictions of different architectures? Are there examples that wide networks are more likely to do well on compared to deep networks, and vice versa?

By training populations of networks on CIFAR-10 and ImageNet, we find that there is considerable diversity in output predictions at the *individual example* level, and broadly, architectures that are more similar in structure have more similar output predictions. On ImageNet we also find that there are statistically significant differences in class-level error rates between wide and deep models, with the former exhibiting a small advantage in identifying classes corresponding to scenes over objects.

Figure 7a compares per-example accuracy for groups of 100 architecturally identical deep models (ResNet-62) and wide models (ResNet-14 (2×)), all trained from different random initializations on CIFAR-10. Although the *average* accuracy of these groups is statistically indistinguishable, they tend to make different errors, and differences between groups are substantially larger than expected

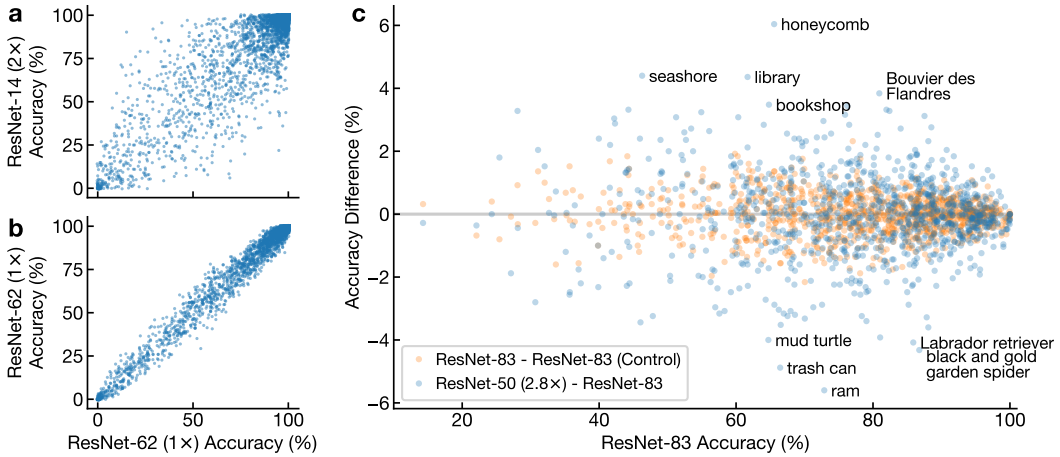

Figure 7: Systematic per-example and per-class performance differences between wide and deep models. **a**: Comparison of accuracy on individual examples for 100 ResNet-62 (1×) and ResNet-14 (2×) models, which have statistically indistinguishable accuracy on the CIFAR-10 test set. **b**: Same as (a), for disjoint sets of 100 architecturally identical ResNet-62 models trained from different initializations. See Figure F.1 for a similar plot for ResNet-14 (2×) models. **c**: Accuracy differences on ImageNet classes for ResNets between models with increased width (y-axis) or depth (x-axis) in the third stage. Orange dots reflect difference between two sets of 50 architecturally identical deep models (i.e., different random initializations of ResNet-83).

by chance (Figure 7b). Examples of images with large accuracy differences are shown in Appendix Figure F.1, while Appendix Figures F.2 and F.3 further explore patterns of example accuracy for networks of different depths and widths, respectively. As the architecture becomes wider or deeper, accuracy on many examples increases, and the effect is most pronounced for examples where smaller networks were often but not always correct. At the same time, there are examples that larger networks are *less* likely to get right than smaller networks. We show similar results for ImageNet networks in Appendix Figure F.4.

We next ask whether wide and deep ImageNet models have systematic differences in accuracy at the class level. As shown in Figure 7c, there are small but statistically significant differences in accuracy for $419/1000$ classes ($p < 0.05$, Welch's t-test), accounting for 11% of the variance in the differences in example-level accuracy (see Appendix F.3). Three of the top 5 classes that are more likely to be correctly classified by wide models reflect scenes rather than objects (seashore, library, bookshop). Indeed, the wide architecture is significantly more accurate on the 68 ImageNet classes descending from "structure" or "geological formation" ($74.9\% \pm 0.05$ vs. $74.6\% \pm 0.06$, $p = 6 \times 10^{-5}$, Welch's t-test). Looking at synsets containing > 50 ImageNet classes, the deep architecture is significantly more accurate on the 62 classes descending from "consumer goods" ($72.4\% \pm 0.07$ vs. $72.1\% \pm 0.06$, $p = 0.001$; Table F.2). In other parts of the hierarchy, differences are smaller; for instance, both models achieve 81.6% accuracy on the 118 dog classes ($p = 0.48$).

8 Conclusion

In this work, we study the effects of width and depth on neural network representations. Through experiments on CIFAR-10, CIFAR-100 and ImageNet, we have demonstrated that as either width or depth increases relative to the size of the dataset, analysis of hidden representations reveals the emergence of a characteristic *block structure* that reflects the similarity of a dominant first principal component, propagated across many network hidden layers. Further analysis finds that while the block structure is unique to each model, other learned features are shared across different initializations and architectures, particularly across relative depths of the network. Despite these similarities in representational properties and performance of wide and deep networks, we nonetheless observe that width and depth have different effects on network predictions at the example and class levels. There remain interesting open questions on how the block structure arises through training, and using the insights on network depth and width to inform optimal task-specific model design.